TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Standardization of Hydrochloric Acid with Secondary Standard Sodium Hydroxide

The chain of standardisation extended one link — the acid certified against the alkali that Experiment 13 certified against oxalic acid.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 14 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to determine the exact concentration of a prepared HCl solution using the NaOH solution standardised in Experiment 13. The NaOH here is a **secondary standard**: it was not weighed into accuracy but *titrated* into it, and it now lends that certified value to the acid. The reaction is the simplest in the course:

$$HCl + NaOH \rightarrow NaCl + H_2O \quad (1\ mol : 1\ mol)$$

Indicator: the strong acid–strong base titration has its equivalence point at pH 7 with a long steep section (Experiment 16), so phenolphthalein (pink → colourless, titrating alkali into acid or the reverse as arranged) or methyl orange both serve; the laboratory's standing choice should be used consistently. In this sheet the acid is pipetted into the flask and the standard alkali run in from the burette, with phenolphthalein disappearing at the endpoint — or methyl orange turning from red to the first permanent orange-yellow, per local practice.

2. Procedure

1. Fill the burette with the standardised NaOH (concentration from Experiment 13, e.g. 0.1016 M), rinsed and bubble-free.
2. Pipette 25.0 mL of the HCl solution into a conical flask; add 2–3 drops of indicator.
3. Titrate with swirling to the endpoint (first permanent colour change), record, and repeat to concordance (within 0.1 mL).
4. Compute from the 1 : 1 stoichiometry; label the acid bottle with the standardised concentration and date.

3. Worked calculation (model data)

STANDARDISATION RESULT

Standard NaOH: 0.1016 M (Experiment 13). HCl pipetted: 25.0 mL. Concordant titre: 23.9 mL.

Moles of NaOH used = $0.1016 \times 0.0239 = 2.428 \times 10^{-3}$ mol.

Moles of HCl in the pipette = the same (1 : 1) = 2.428×10^{-3} mol.

Molarity of HCl = $2.428 \times 10^{-3} / 0.0250$ = **0.0971 M**.

The acid was “prepared as 0.1 M” (Experiment 12); its certified value is 0.0971 M, and 0.0971 M is what every later user of the bottle must calculate with.

4. Why a secondary standard is acceptable — and its limit

Analytical chains are built this way everywhere: primary standard → secondary standard → working solution. Each link adds a small titration uncertainty, so the chain is kept **short**, the links **recent** (restandardise NaOH that has stood open to air), and critical work goes back to a primary standard directly — as Experiment 15 does for this same acid, giving the course two independent values for one bottle, whose agreement is the real lesson.

5. Precautions

- The NaOH burette solution absorbs CO_2 while it stands in the burette; do not leave it there between practical classes, and never return unused burette solution to the stock bottle.
- Use the same indicator and endpoint colour as in any comparison titration, so systematic indicator error cancels between experiments.
- Record the Experiment 13 value used, with its date, in this experiment’s write-up — a standardisation without provenance is an orphan number.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Explain “secondary standard” in one sentence, using this experiment as the example.
2. 25.0 mL of HCl requires 24.4 mL of 0.1016 M NaOH. Find the acid’s molarity.
3. Why may either phenolphthalein or methyl orange be used here, when only phenolphthalein served in Experiment 13?
4. The HCl was “prepared as 0.1 M” yet standardises at 0.0971 M. Give two causes that can produce such a shortfall.

5. Why should standardisation chains be kept short?
6. The standardised NaOH has stood in an open bottle for a month. What has happened to it, and what must be done before it standardises anything else?

Answer key

1. A secondary standard is a solution whose concentration is known not by weighing a primary substance but by titration against one — here the NaOH, certified in Experiment 13 against oxalic acid, certifies the HCl in its turn.
2. Moles NaOH = $0.1016 \times 0.0244 = 2.479 \times 10^{-3}$; equal moles of HCl in 25.0 mL → M = $2.479 \times 10^{-3}/0.0250 = 0.0992$ M.
3. Because the strong–strong equivalence is at pH 7 with a steep section spanning both indicators' ranges (about pH 4–10), so either changes within a drop of equivalence. Experiment 13's weak acid put its equivalence near pH 8–9, inside phenolphthalein's range only.
4. The concentrate's true strength differed from the assumed 37%/12.1 M (bottle variation, HCl gas lost on repeated opening), and dilution/measurement tolerances in Experiment 12. The prepared value is nominal by construction — which is why this experiment exists.
5. Each titration link contributes its own small error (endpoint, reading, indicator), and errors accumulate along the chain; short chains keep the final value close to the primary standard's certainty.
6. It has absorbed CO_2 (and water vapour exchange), converting some NaOH to Na_2CO_3 and lowering its effective alkali strength — its certified value has expired. It must be restandardised against the primary standard (Experiment 13) before further use.

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